



December 2025

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from Unit – I & Unit – II each and **One** full question from Unit – III.

10. The critical-section problem could be solved in a single-core environment by preventing _____ from occurring
- A) interrupts B) processes
C) requests D) resources
11. A _____ kernel is essentially free from race conditions
- A) preemptive B) Supervisor
C) user D) Non-preemptive
12. A _____ is an ADT that includes a set of programmer-defined operations that are provided with mutual exclusion within the monitor
- A) Critical type B) Monitor type
C) Exec type D) Wait type
13. The _____ is used as an index into the page table.
- A) frame bit B) Frame offset
C) Page number D) Page offset
14. With paging there is no _____ fragmentation.
- A) internal B) external
C) Either type D) Concept of
15. If the process can be moved during its execution from one memory segment to another, then binding must be _____
- A) delayed until run time B) preponed to compile time
C) preponed to load time D) preponed to run time
16. Physical memory is broken into fixed-sized blocks called _____
- A) Pages B) Blocks
C) Segments D) Frames
17. When the page fault rate is low _____
- A) the turnaround time increases B) turnaround time & effective access time increases
C) the effective access time decreases D) the effective access time increases
18. The data structure used for file directory is called _____
- A) mount table B) Process table
C) file table D) Hash table
19. Which free space management method maintains a chain of free disk blocks and allocates blocks from the chain?
- A) Bit Map B) Linked List
C) Indexed Allocation D) Contiguous Allocation
20. The directory can be viewed as a _____ that translates file names into their directory entries.
- A) Symbol table B) partition
C) Cache D) Swap space

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I****Marks BT* CO* PO***

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|---|---|-----|---|---|
| 1. a) Explain the main services provided by an operating system with suitable examples. | 6 | L*2 | 1 | 1 |
| b) Outline the role of linker and loaders in program execution. | 6 | L2 | 1 | 1 |
| c) Distinguish between user mode and kernel mode. | 4 | L4 | 1 | 1 |
| 2. a) Explain the scheduling criteria used to evaluate CPU scheduling algorithms. | 6 | L2 | 2 | 1 |

- b) The following processes are being scheduled using a preemptive, round robin scheduling algorithm.

Process	Priority	Burst time	Arrival time
P1	40	20	0
P2	30	25	25
P3	30	25	30
P4	35	15	60
P5	5	15	100
P6	10	10	105

Each process is assigned a numerical priority, with a higher number indicating a higher relative priority. In addition to the processes listed below, the system also has an **idle task** (which consumes no CPU resources and is identified as *Pidle*). This task has priority 0 and is scheduled whenever the system has no other available processes to run. The length of a time quantum is 10 units. If a process is re-empted by a higher-priority process, the re-empted process is placed at the end of the queue.

- I. Show the scheduling order of the processes using a Gantt chart.
- II. What is the turnaround time for each process?
- III. What is the waiting time for each process?
- IV. What is the CPU utilization ?

10 L3 2 2

3. a) Distinguish between multithreading models.

6 L4 2 1

- b) A CPU executes processes using time quanta (round-robin). Each context switch takes 5 μ s. Each process gets a time quantum of 50 μ s.

i. What fraction of CPU time is wasted in context switching?

ii. What is the CPU utilization for useful work?

4 L3 2 2

- c) The performance of the Round Robin algorithm depends heavily on the size of the time quantum. Justify.

6 L4 2 2

Unit – II

4. a) Outline the necessary conditions for a deadlock to occur.
 b) Explain Peterson's Solution for two processes with its algorithm and working.
 c) Explain deadlock recovery.

4 L2 3 1

6 L3 3 1

6 L2 3 1

5. a) What is the Critical Section Problem? State the requirements that any solution to this problem must satisfy.
 b) With a diagram, explain the mapping of logical and physical address spaces.
 c) Describe the classical Producer-Consumer problem and show how semaphores solve it.

6 L2 3 1

6 L2 4 1

4 L3 3 1

6. a) With the help of a neat diagram, describe the Hierarchical Paging Table.

6 L2 4 1

- b) A system has 3 types of resources

A B C

7 5 7

Process	Max			Allocation		
	A	B	C	A	B	C
P1	6	5	3	1	0	2
P2	4	2	1	3	1	0
P3	5	1	2	0	1	1
P4	1	0	2	1	0	1
P5	5	2	3	1	0	2

Determine whether the system is safe.

10 L3 3 2

Unit – III

7. a) With the help of a diagram, explain how a page fault is handled .
 b) Compare the performance of FIFO, LRU and Optimal page replacement algorithms with 3 frames. The reference string is 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2.

6 L2 4 1

10 L3 4 1

8. a) Outline the advantages and disadvantages of different file allocation methods.
 b) Compute total head movement for SSTF, SCAN, and C-SCAN and algorithms.

6 L4 5 1

Initial head at 50. Request queue : 82, 170, 43, 140, 24, 16, 190.

Assume disk head moves toward higher numbers first.

10 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
